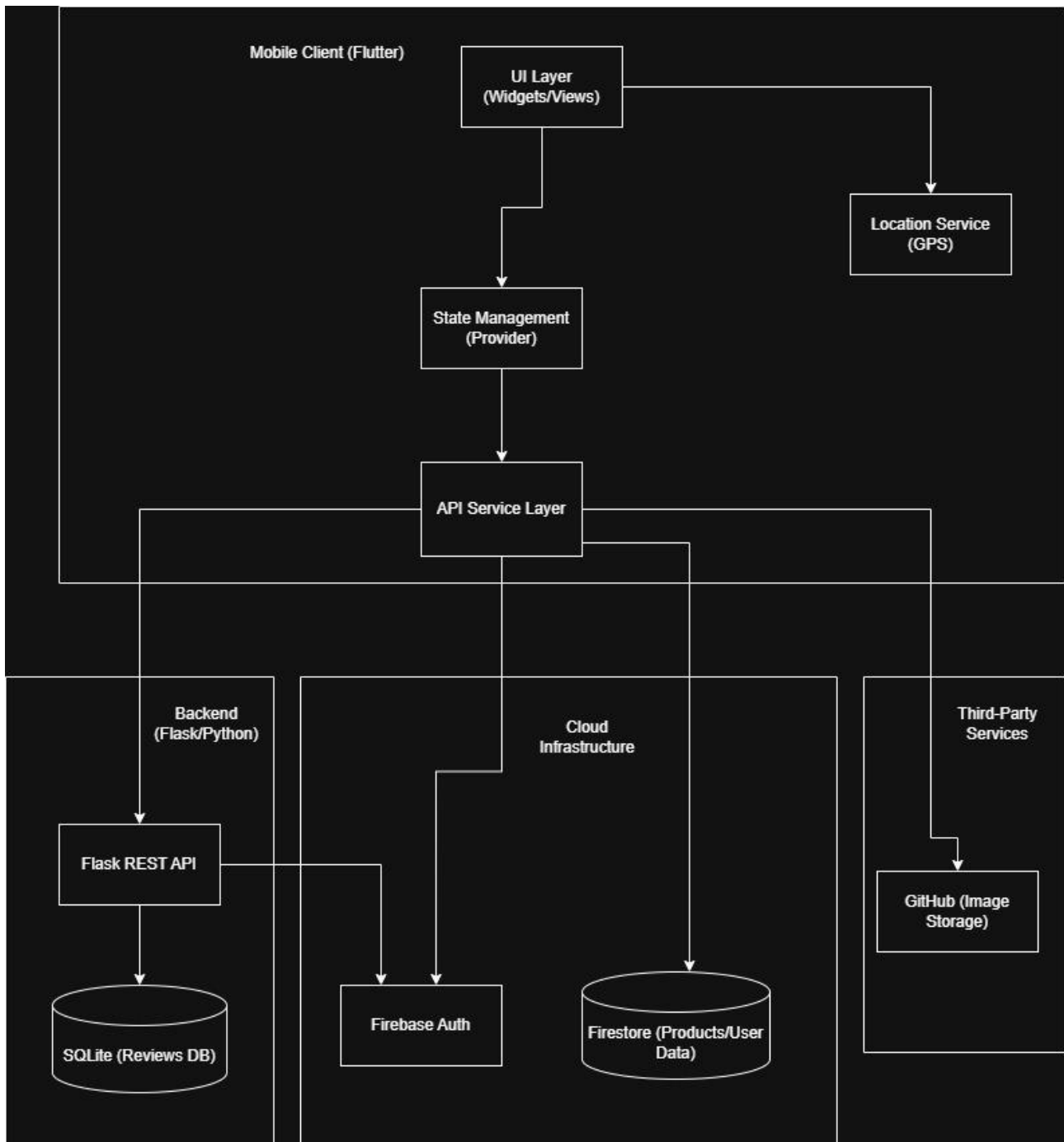


APPLICATION ARCHITECTURE



Description

1. Flutter Mobile Application (Frontend)

- UI Layer: modular widgets and screens following Material Design;
- Provider: manage data like shopping cart and user favourites;

- Centralized services like ReviewService and GitStorageService, handle HTTP requests and data mapping;
- Location Services: Uses device GPS to calculate proximity between users and shops for real-time product filtering.

2. Flask REST API (Backend)

- Endpoints: provides CRUD operations for product reviews;
- Security: Verifies Firebase ID Tokens (JWT) sent from the mobile app to ensure only authenticated users can post/edit reviews;
- Storage: Uses at-file SQLite for lightweight, persistent storage of review metadata.

3. Cloud Infrastructure

- Firebase Authentication: Manages secure user login (Email/Password, Google);
- Google Firestore: A NoSQL database storing the primary product catalog, shop details, and user profiles.

4. Third Party Services

- GitHub API: Leverages GitHub as a Content Delivery Network (CDN) for product images, managed via the GitStorageService.

NAVIGATION GRAPH

